

Session 22

JavaScript Arrays and Core Array Operations

What are you going to learn in this session?

In this session, you will learn how JavaScript uses arrays to store and manage collections of data efficiently. Arrays are one of the most important data structures in programming and are widely used in real-world applications such as storing user lists, product data, API responses, and dynamic content. You will understand how arrays work internally, how indexing allows access to elements, and how data is organized sequentially.

You will also learn how to use the `for...of` loop to iterate through arrays in a clean and readable way. Along with iteration, you will explore essential array methods such as `push`, `pop`, `shift`, `unshift`, `slice`, `splice`, `reverse`, `join`, and `toString`. These methods help in adding, removing, modifying, and transforming array data.

By the end of this session, you will be able to confidently manipulate arrays, perform common data operations, and build real-world features like lists, menus, carts, and dynamic datasets used in web applications.

Why is this topic important?

Arrays are the backbone of data handling in software development. Every application that deals with multiple records—such as e-commerce platforms, social media feeds, dashboards, and APIs—relies heavily on arrays. Understanding array operations allows developers to efficiently manage and transform large sets of data in a structured way.

Interviewers frequently test arrays because they reflect a candidate's problem-solving skills and understanding of data structures. Concepts like indexing, iteration, and array manipulation are essential for solving coding problems, building algorithms, and working with frameworks like React and Node.js. Strong knowledge of arrays is critical for both frontend and backend development roles.

How should you prepare yourself to attend this session?

Before attending this session, you should be comfortable with JavaScript fundamentals such as variables, loops, functions, and conditional statements. Since arrays involve iteration and data manipulation, understanding `for` loops and `for...of` loops from previous sessions will be very helpful.

Ensure that Node.js and VS Code are properly set up on your system. Practice basic JavaScript programs involving loops and conditions. Revisiting concepts like data types, objects, and functions will help you understand how arrays interact with other structures in real applications.

What should you do after the session?

After completing this session, practice working with arrays daily. Start with simple tasks such as adding and removing elements, looping through arrays, and transforming data. Gradually move to building small projects like to-do lists, product catalogs, and student management systems.

You should also revise array methods regularly and try combining multiple methods in a single program. For interview preparation, focus on explaining differences between methods like `slice` vs `splice` and practicing coding problems based on array manipulation. This will strengthen your logical thinking and problem-solving skills.

Class Notes

Array Basics and Indexing

An array is a special variable used to store multiple values in a single structure. Each value in an array has an index starting from 0.

Example

JavaScript

```
let fruits = ["Apple", "Banana", "Mango"];
```

```
console.log(fruits[0]);
```

Output

None

Apple

Key Points

- Index starts from 0
- Can store multiple data types
- Dynamic in size

Real-World Use Cases

- Product lists
- User data
- API responses
- Shopping carts

for...of Loop

The `for...of` loop is used to iterate over array values directly.

Example

JavaScript

```
let numbers = [10, 20, 30];

for (let num of numbers) {
  console.log(num);
}
```

Output

None

10

20

30

Benefits

- Cleaner syntax than traditional loops
- Easy to read
- Best for arrays

push()

Adds an element at the end of an array.

Example

JavaScript

```
let arr = [1, 2];
```

```
arr.push(3);
```

```
console.log(arr);
```

Output

None

```
[1, 2, 3]
```

pop()

Removes the last element from an array.

Example

JavaScript

```
let arr = [1, 2, 3];  
  
arr.pop();  
  
console.log(arr);
```

Output

None

```
[1, 2]
```

shift()

Removes the first element of an array.

Example

JavaScript

```
let arr = [1, 2, 3];  
  
arr.shift();  
  
console.log(arr);
```

Output

None

[2, 3]

unshift()

Adds an element at the beginning of an array.

Example

JavaScript

```
let arr = [2, 3];
```

```
arr.unshift(1);
```

```
console.log(arr);
```

Output

None

[1, 2, 3]

slice()

Returns a portion of an array without modifying the original array.

Example

JavaScript

```
let arr = [1, 2, 3, 4];
```

```
let newArr = arr.slice(1, 3);
```

```
console.log(newArr);
```

Output

None

[2, 3]

Key Point

- Does NOT change original array

splice()

Used to add or remove elements from an array (modifies original array).

Example

JavaScript

```
let arr = [1, 2, 3, 4];
```

```
arr.splice(1, 2);
```

```
console.log(arr);
```

Output

None

[1, 4]

reverse()

Reverses the order of elements in an array.

Example

JavaScript

```
let arr = [1, 2, 3];
```

```
arr.reverse();
```

```
console.log(arr);
```

Output

None

[3, 2, 1]

join()

Converts array elements into a single string.

Example

JavaScript

```
let arr = ["A", "B", "C"];
```



```
console.log(arr.join("-"));
```

Output

None

A-B-C

toString()

Converts array into a comma-separated string.

Example

JavaScript

```
let arr = [1, 2, 3];
```

```
console.log(arr.toString());
```

Output

None

1,2,3

delete operator

Removes an element but leaves an empty slot in the array.

Example

JavaScript

```
let arr = [1, 2, 3];
```

```
delete arr[1];
```

```
console.log(arr);
```

Output

None

```
[1, empty, 3]
```

Key Point

- Not recommended for arrays
- Use `splice()` instead

Interview Prep Recap

Students should now be able to confidently explain:

- What arrays are and how indexing works
- How to iterate arrays using `for...of`
- Differences between array methods
- When to use push, pop, shift, unshift
- Difference between slice and splice
- How reverse, join, and toString work
- Why delete is not recommended for arrays
- Real-world use of arrays in applications

Sample Interview Questions

Q1. What is an array?

Answer:

An array is a collection of multiple values stored in a single variable.

Q2. Difference between slice and splice?

Answer:

slice returns a new array without modifying original, splice modifies original array.

Q3. What does push do?

Answer:

Adds an element at the end of an array.

Q4. Why is delete not recommended for arrays?

Answer:

Because it leaves empty slots instead of shifting elements.

Q5. What is for...of loop used for?

Answer:

To iterate through array values directly.

Common Beginner Confusions

Confusion:1

Array and object are the same.

Reality

Arrays store ordered values; objects store key-value pairs.

How to Remember It

Array = list, Object = dictionary.

Confusion:2

push and unshift do the same thing.

Reality

push adds to end, unshift adds to beginning.

How to Remember It

push = back, unshift = front.

Confusion:3

splice and slice are interchangeable.

Reality

slice does not modify array, splice modifies array.

How to Remember It

slice = safe copy, splice = modify.

Confusion:4

delete removes elements properly.

Reality

delete leaves empty slots.

How to Remember It

delete = remove value, not structure.

Confusion:5

for loop and for...of are identical.

Reality

for...of directly iterates values.

How to Remember It

for = control, for...of = direct values.

Confusion:6

reverse creates a new array.

Reality

reverse modifies original array.

How to Remember It

reverse = in-place change.

Confusion:7

join and toString are the same.

Reality

join allows custom separator.

How to Remember It

join = flexible, toString = fixed comma.

Confusion:8

Arrays cannot store multiple types.

Reality

JavaScript arrays can store mixed types.

How to Remember It

JS array = flexible container.

Confusion:9

Index starts from 1.

Reality

Index starts from 0.

How to Remember It

First item is always 0.

Key Takeaways

In this session, you learned how arrays are used to store and manage multiple values efficiently in JavaScript. You explored indexing, iteration using `for...of`, and essential array operations such as adding, removing, and modifying elements. These concepts are fundamental for handling structured data in real-world applications.

You also learned powerful array methods like `push`, `pop`, `shift`, `unshift`, `slice`, `splice`, `reverse`, `join`, and `toString`, each serving a specific purpose in data manipulation. Understanding when and how to use these methods helps you write cleaner, more efficient code and solve practical programming problems.

From an industry perspective, arrays are used everywhere—from frontend UI rendering to backend data processing and APIs. Mastering arrays is essential for coding interviews, algorithm challenges, and building scalable applications. The next step is to combine arrays with functions and objects to build more complex real-world systems.